

6260 Final Project Report

Emily Wu and Cameron Barlow

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Motivation

Is there a relationship between early life socioeconomic characteristics and family background that can predict the severity of an adult's criminal history? We could create the response of criminal history using an ordinal scale to predict the severity part of criminal history using a cumulative logit model.

Data

[Talk about sampling] [Talk about how much data we have available]

The response variable was created through the synthesis of maximum sentence and crime type to model severity. The time variable is highly continuous and skewed in raw form, but much more regular after log transformation, so if ordinal categories are used, they should be created deliberately for interpretability or modeling convenience.

```

#proposed variables
varlist<-c("V0062", "V0411", "V0413", "V0414", "V0415", "V0935", "V0944", "V0945", "V0982", "V1166", "V1165")
varlist<- c(varlist, v_range("1170", "1174"))

sblast<- data%>%
  select(all_of(varlist))

sblast <- sblast %>%
  rename(offense_num = V0062) %>%
  rename("Years of Education" = V0935) %>%
  rename("Total Max Sentence" = V0411) %>%
  rename("Max Sentence # of Years" = V0413) %>%
  rename("Max Sentence # of Months" = V0414) %>%
  rename("Max Sentence # of Days" = V0415) %>%
  rename("SpEd" = V0944) %>%
  rename("Born in the US" = V0945) %>%
  rename("Children" = V0982) %>%
  rename("Foster Home" = V1166) %>%
  rename("Homeless" = V1165) %>%
  rename("Specify Home or Institution" = V1167) %>%
  rename("Lived with most time" = V1168) %>%
  rename("Welfare" = V1170) %>%
  rename("Public Housing" = V1171) %>%
  rename("Father" = V1172) %>%
  rename("Mother" = V1173) %>%
  rename("Sibling" = V1174)
sblast <- sblast %>%
  mutate(across(everything(), ~ as.numeric(gsub(".*\\((-?\\d+)\\).*", "\\1", as.character(.x))))) #reference

sblast$Tot_time <- (sblast$`Max Sentence # of Years`*365 + sblast$`Max Sentence # of Months`*30 + sblast$`Max Sentence # of Days`)
sblast <- sblast %>%
  mutate(edu_group = case_when(
    `Years of Education` < 12 ~ "< HS",
    `Years of Education` == 12 ~ "HS/GED",
    `Years of Education` > 12 & `Years of Education` < 16 ~ "Some college",
    `Years of Education` >= 16 ~ "College+"
  ))
sblast <- sblast %>%
  mutate(offense_type = case_when(
    offense_num == 1 ~ "Violent",
    offense_num == 2 ~ "Property",
    offense_num == 3 ~ "Drug",
    offense_num == 4 ~ "Public Order",
    offense_num == 5 ~ "Other",
    TRUE ~ "Unknown"
  ))

sblast2<-sblast[!is.na(sblast$Tot_time),] #only where data is available

```

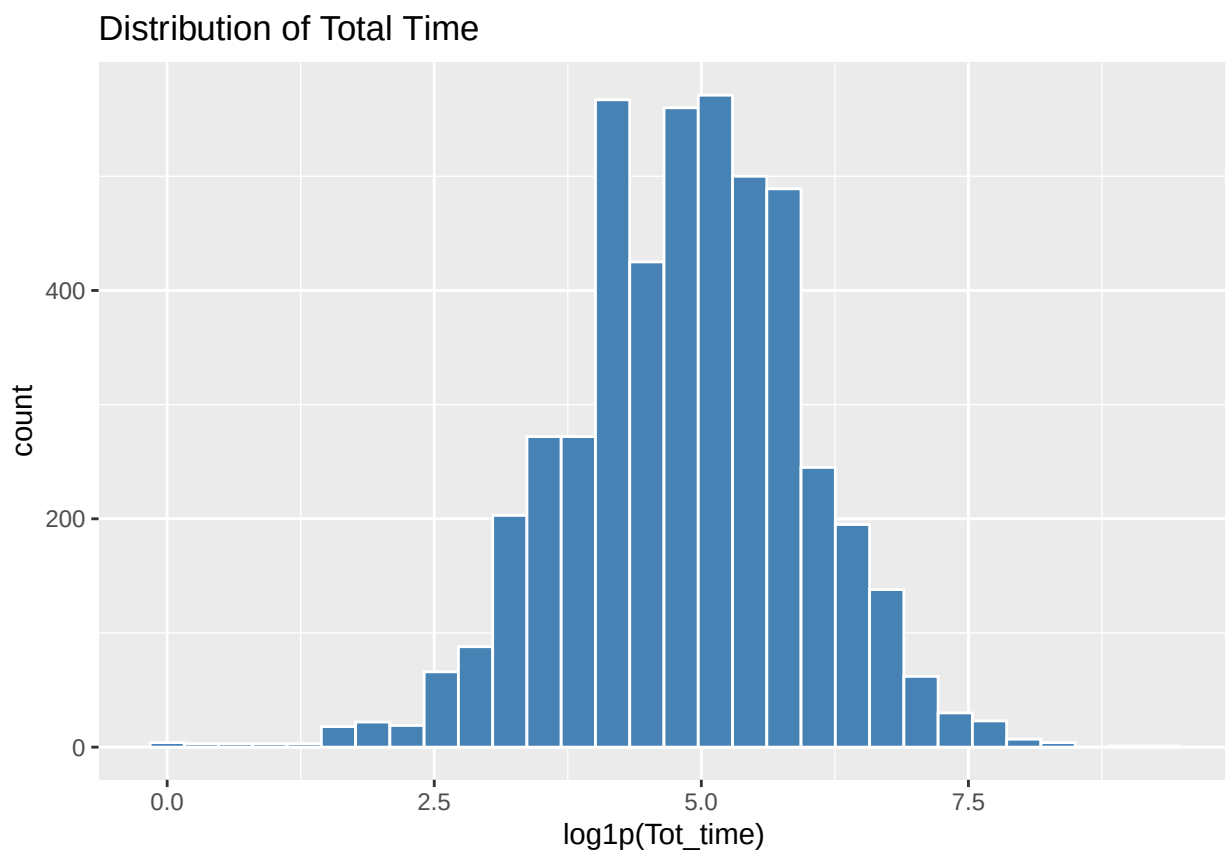
#heavily right skewed

```
ggplot(sblast, aes(x = log1p(Tot_time))) +  
  geom_histogram(fill = "steelblue", color = "white") +  
  labs(title = "Distribution of Total Time")
```

Warning in log1p(Tot_time): NaNs produced

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

Warning: Removed 15270 rows containing non-finite outside the scale range
('stat_bin()').

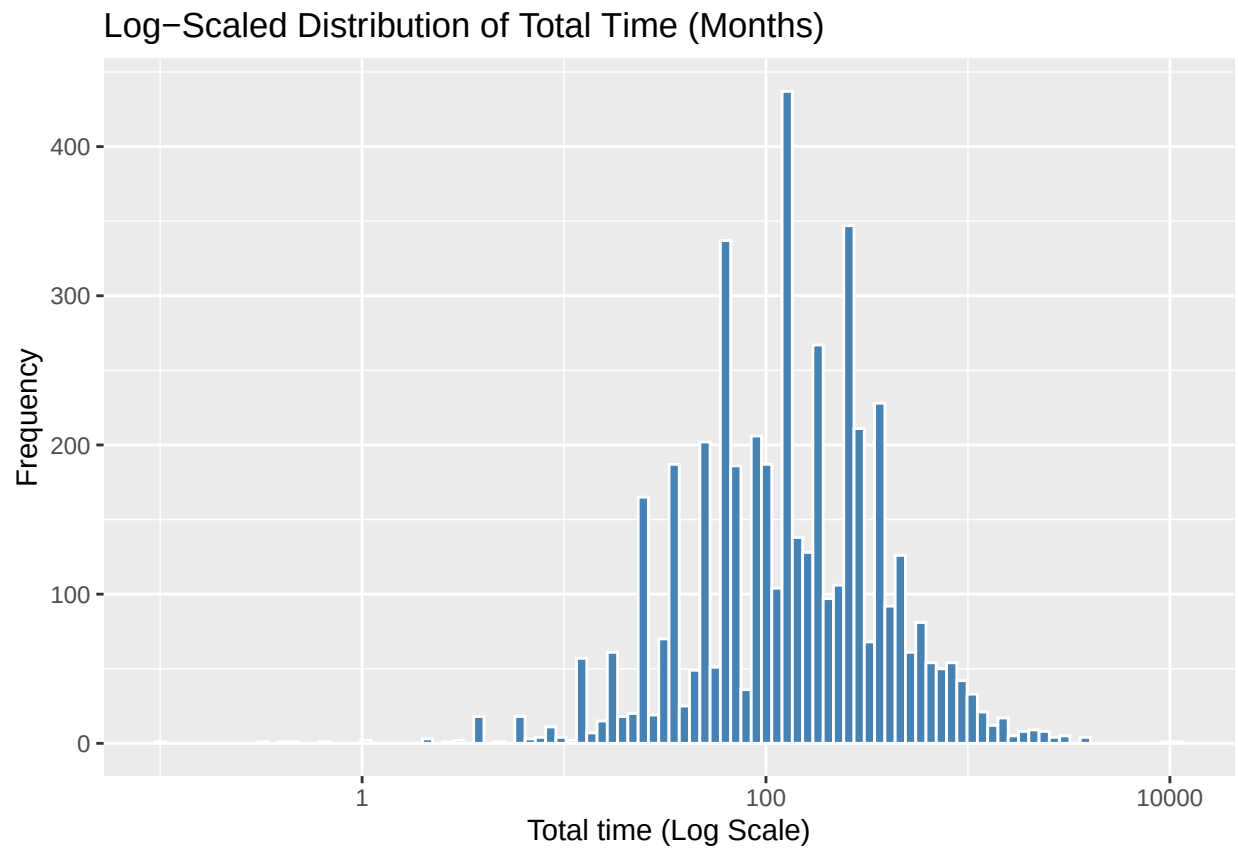


```
ggplot(sblast, aes(x = Tot_time)) +  
  geom_histogram(bins = 100, fill = "steelblue", color = "white") +  
  scale_x_log10() +  
  labs(title = "Log-Scaled Distribution of Total Time (Months)",  
       x = "Total time (Log Scale)",  
       y = "Frequency")
```

Warning in transformation\$transform(x): NaNs produced

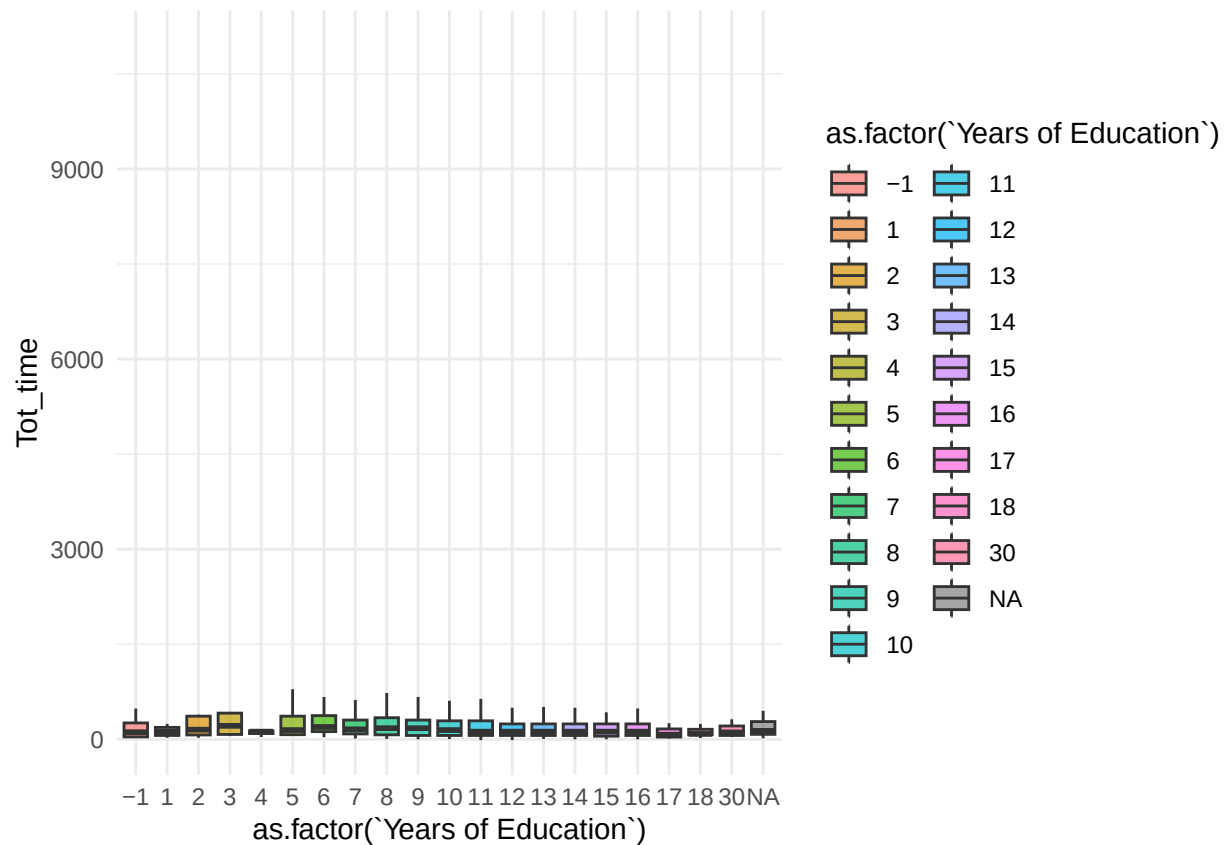
Warning in scale_x_log10(): log-10 transformation introduced infinite values.

```
## Warning: Removed 15273 rows containing non-finite outside the scale range
## ('stat_bin()').
```



Education Eda/sh

```
ggplot(sblast2, aes(x = as.factor(`Years of Education`), y = Tot_time, fill = as.factor(`Years of Education`))) +  
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +  
  theme_minimal()
```



```
#scatterplot with log-transformed prison time
ggplot(sblast2, aes(x = edu_group, y = log1p(Tot_time))) +
  geom_jitter(width = 0.2, height = 0.05, alpha = 0.4) +
  geom_smooth(method = "loess", se = TRUE) +
  labs(
    x = "Years of Education",
    y = "log(1 + Total Time in Prison)",
    title = "Education and Total Time in Prison"
  ) +
  theme_minimal()
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

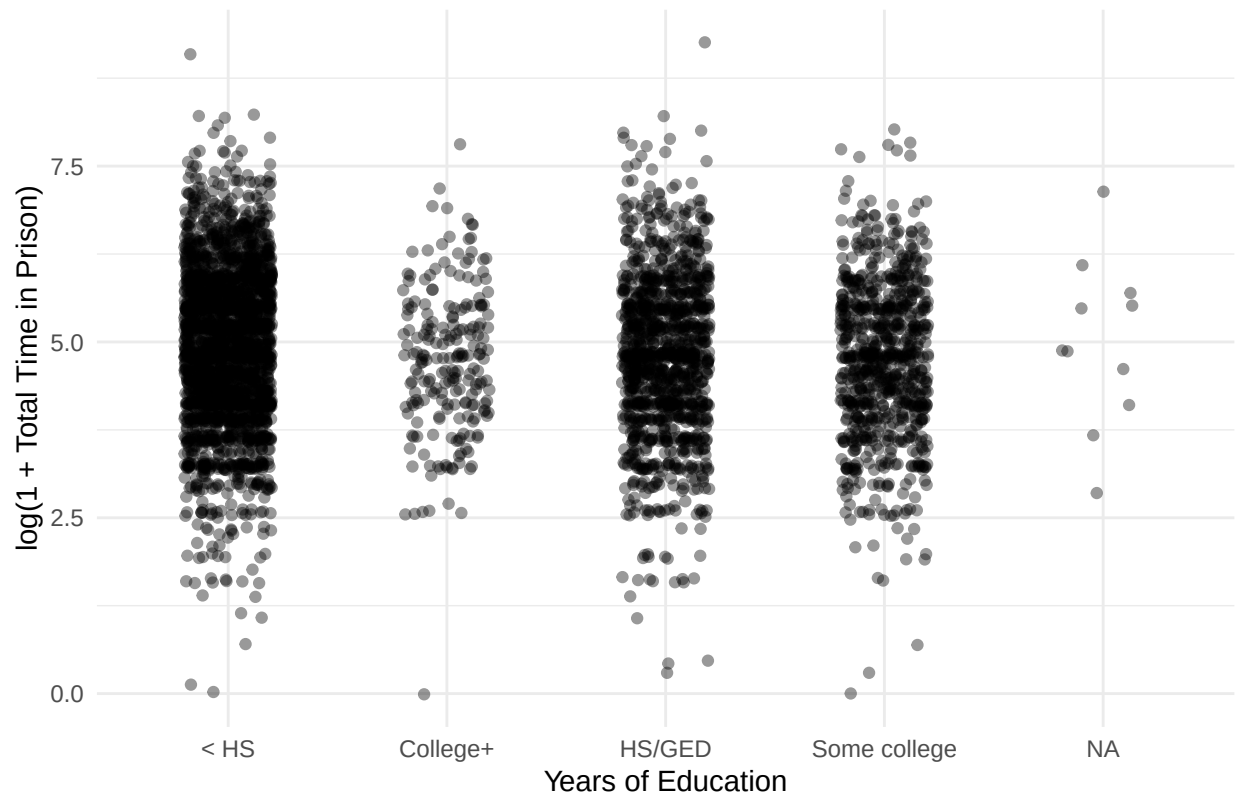
```
## Warning: Removed 3 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning: Removed 3 rows containing missing values or values outside the scale range
```

```
## ('geom_point()').
```

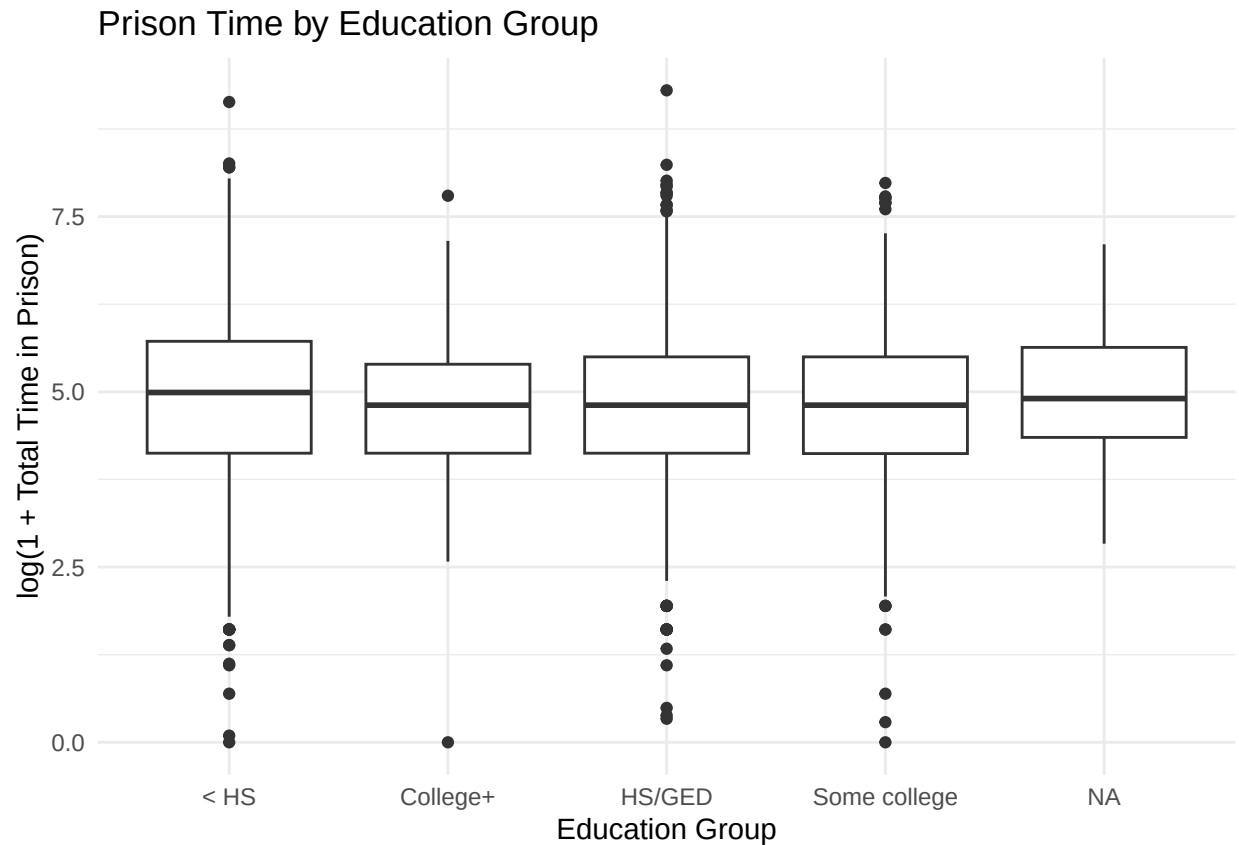
Education and Total Time in Prison



```
ggplot(sblast, aes(x = edu_group, y = log1p(Tot_time))) +
  geom_boxplot() +
  labs(
    x = "Education Group",
    y = "log(1 + Total Time in Prison)",
    title = "Prison Time by Education Group"
  ) +
  theme_minimal()
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## Warning: Removed 15270 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```

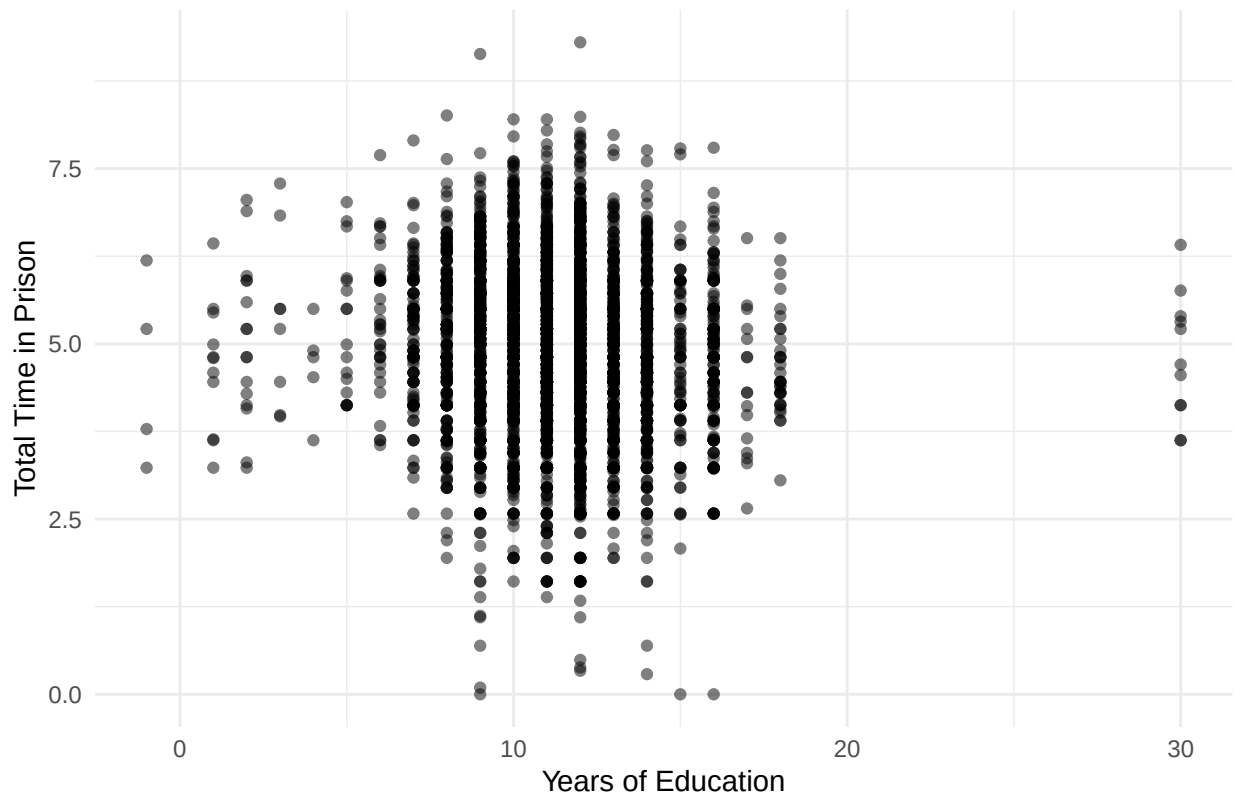



```
ggplot(sblast2, aes(x = `Years of Education`, y = log1p(Tot_time))) +
  geom_point(alpha = 0.5) +
  labs(
    x = "Years of Education",
    y = "Total Time in Prison",
    title = "Years of Education vs Total Time in Prison"
  ) +
  theme_minimal()
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Years of Education vs Total Time in Prison



```
sblast2[sblast2$SpEd<0,] #20 observations missing for those who were SpEd or not
```

```
##      offense_num Total Max Sentence Max Sentence # of Years
## 213           1           1           1           12
## 1564          1           1           1          235
## 2241          2           1           1           0
## 4076          2           1           1           5
## 4920          1           1           1          55
## 5421          4           1           1           0
## NA           NA           NA           NA          NA
## NA.1          NA           NA           NA          NA
## NA.2          NA           NA           NA          NA
## 9821          2           1           1           6
## NA.3          NA           NA           NA          NA
## 10938         4           1           1          12
## NA.4          NA           NA           NA          NA
## NA.5          NA           NA           NA          NA
## NA.6          NA           NA           NA          NA
## NA.7          NA           NA           NA          NA
## 15355         3           1           1         165
## NA.8          NA           NA           NA          NA
## NA.9          NA           NA           NA          NA
## NA.10         NA           NA           NA          NA
##      Max Sentence # of Months Max Sentence # of Days Years of Education SpEd
## 213              0              0              10      -1
```

## 1564	0	0	10	-2				
## 2241	46	0	13	-1				
## 4076	0	0	12	-1				
## 4920	0	0	12	-1				
## 5421	27	0	14	-1				
## NA	NA	NA	NA	NA				
## NA.1	NA	NA	NA	NA				
## NA.2	NA	NA	NA	NA				
## 9821	0	0	12	-1				
## NA.3	NA	NA	NA	NA				
## 10938	0	0	11	-1				
## NA.4	NA	NA	NA	NA				
## NA.5	NA	NA	NA	NA				
## NA.6	NA	NA	NA	NA				
## NA.7	NA	NA	NA	NA				
## 15355	0	0	14	-1				
## NA.8	NA	NA	NA	NA				
## NA.9	NA	NA	NA	NA				
## NA.10	NA	NA	NA	NA				
##	Born in the US Children Foster Home Homeless Specify Home or Institution							
## 213	1	1	2	1 NA				
## 1564	1	2	1	1 3				
## 2241	1	2	2	2 NA				
## 4076	1	2	2	2 NA				
## 4920	1	1	2	2 NA				
## 5421	1	1	2	2 NA				
## NA	NA	NA	NA	NA NA				
## NA.1	NA	NA	NA	NA NA				
## NA.2	NA	NA	NA	NA NA				
## 9821	1	1	2	2 NA				
## NA.3	NA	NA	NA	NA NA				
## 10938	1	1	2	1 NA				
## NA.4	NA	NA	NA	NA NA				
## NA.5	NA	NA	NA	NA NA				
## NA.6	NA	NA	NA	NA NA				
## NA.7	NA	NA	NA	NA NA				
## 15355	1	1	2	2 NA				
## NA.8	NA	NA	NA	NA NA				
## NA.9	NA	NA	NA	NA NA				
## NA.10	NA	NA	NA	NA NA				
##	Lived with most time V1264 V1252 Welfare Public Housing Father Mother							
## 213	1	7	18	1	1	1	2	
## 1564	2	10	15	-1	2	-1	1	
## 2241	1	16	16	2	2	2	2	
## 4076	6	10	10	2	2	2	2	
## 4920	1	11	15	2	2	2	2	
## 5421	2	20	NA	1	1	2	2	
## NA	NA	NA	NA	NA	NA	NA	NA	
## NA.1	NA	NA	NA	NA	NA	NA	NA	
## NA.2	NA	NA	NA	NA	NA	NA	NA	
## 9821	2	20	18	2	2	2	2	
## NA.3	NA	NA	NA	NA	NA	NA	NA	
## 10938	2	21	33	1	2	1	2	
## NA.4	NA	NA	NA	NA	NA	NA	NA	

```
## NA.5          NA    NA    NA    NA          NA    NA    NA
## NA.6          NA    NA    NA    NA          NA    NA    NA
## NA.7          NA    NA    NA    NA          NA    NA    NA
## 15355         2    20    NA    1          1     2     2
## NA.8          NA    NA    NA    NA          NA    NA    NA
## NA.9          NA    NA    NA    NA          NA    NA    NA
## NA.10         NA    NA    NA    NA          NA    NA    NA
##      Sibling  Tot_time    edu_group offense_type
## 213         1  146.00000      < HS    Violent
## 1564        1 2859.16667      < HS    Violent
## 2241        2   46.00000 Some college Property
## 4076        1   60.83333    HS/GED    Property
## 4920        1  669.16667    HS/GED    Violent
## 5421        2   27.00000 Some college Public Order
## NA         NA      NA      <NA>      <NA>
## NA.1        NA      NA      <NA>      <NA>
## NA.2        NA      NA      <NA>      <NA>
## 9821        2   73.00000    HS/GED    Property
## NA.3        NA      NA      <NA>      <NA>
## 10938       1  146.00000      < HS Public Order
## NA.4        NA      NA      <NA>      <NA>
## NA.5        NA      NA      <NA>      <NA>
## NA.6        NA      NA      <NA>      <NA>
## NA.7        NA      NA      <NA>      <NA>
## 15355       1 2007.50000 Some college Drug
## NA.8        NA      NA      <NA>      <NA>
## NA.9        NA      NA      <NA>      <NA>
## NA.10       NA      NA      <NA>      <NA>
```

```
ggplot(sblast2, aes(x = SpEd, y = log1p(Tot_time))) +
  geom_jitter(width = 0.2, height = 0.05, alpha = 0.4) +
  geom_smooth(method = "loess", se = TRUE) +
  labs(
    x = "Special Education",
    y = "log(1 + Total Time in Prison)",
    title = "Special Education Status and Total Time in Prison"
  ) +
  theme_minimal()
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 14 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
```

```
## : at 2.02
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
```

```
## : radius 0.0004
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : all data on boundary of neighborhood. make span bigger

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : pseudoinverse used at 2.02

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : neighborhood radius 0.02

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : reciprocal condition number 0.0070342

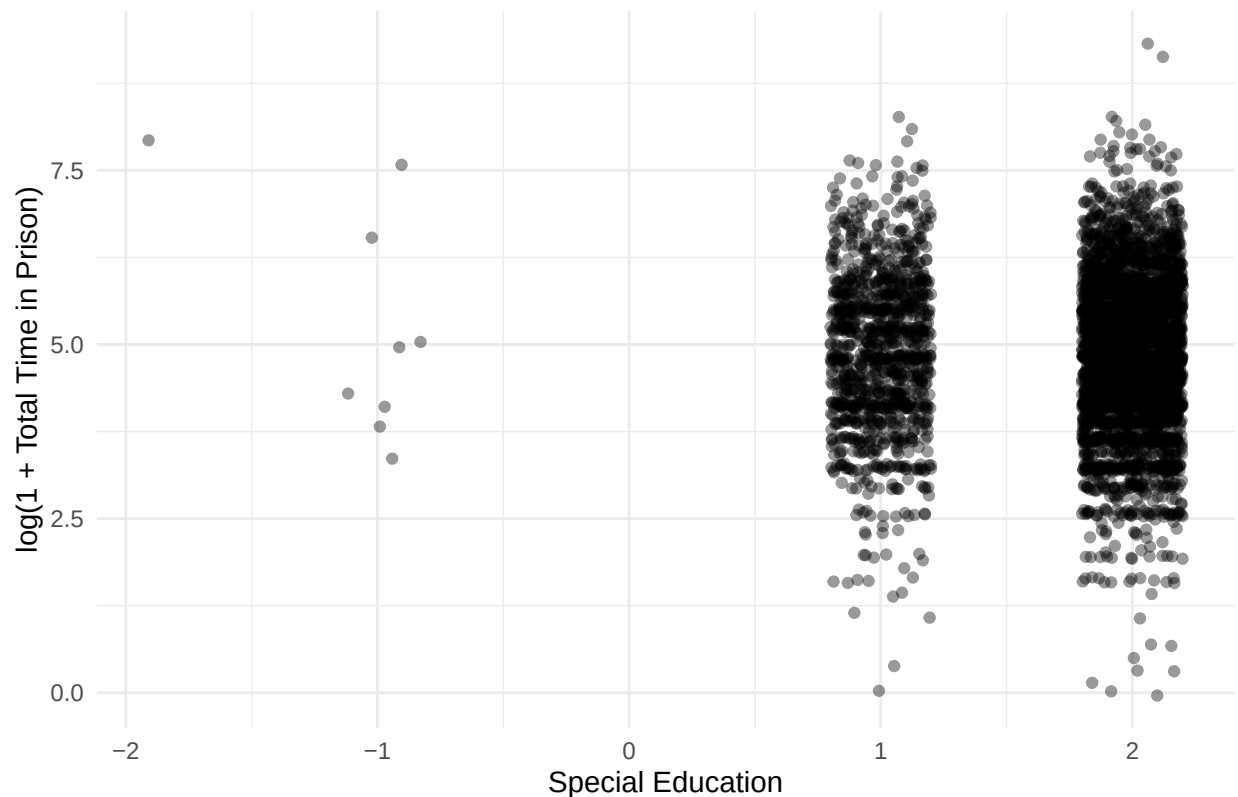
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : There are other near singularities as well. 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : zero-width neighborhood. make span bigger

## Warning: Failed to fit group -1.
## Caused by error in 'predLoess()':
## ! NA/NaN/Inf in foreign function call (arg 5)

## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_point()').
```

Special Education Status and Total Time in Prison



```
sblast2[sblast2$`Born in the US`<0,] #11 observations missing for those who were born in the US or not
```

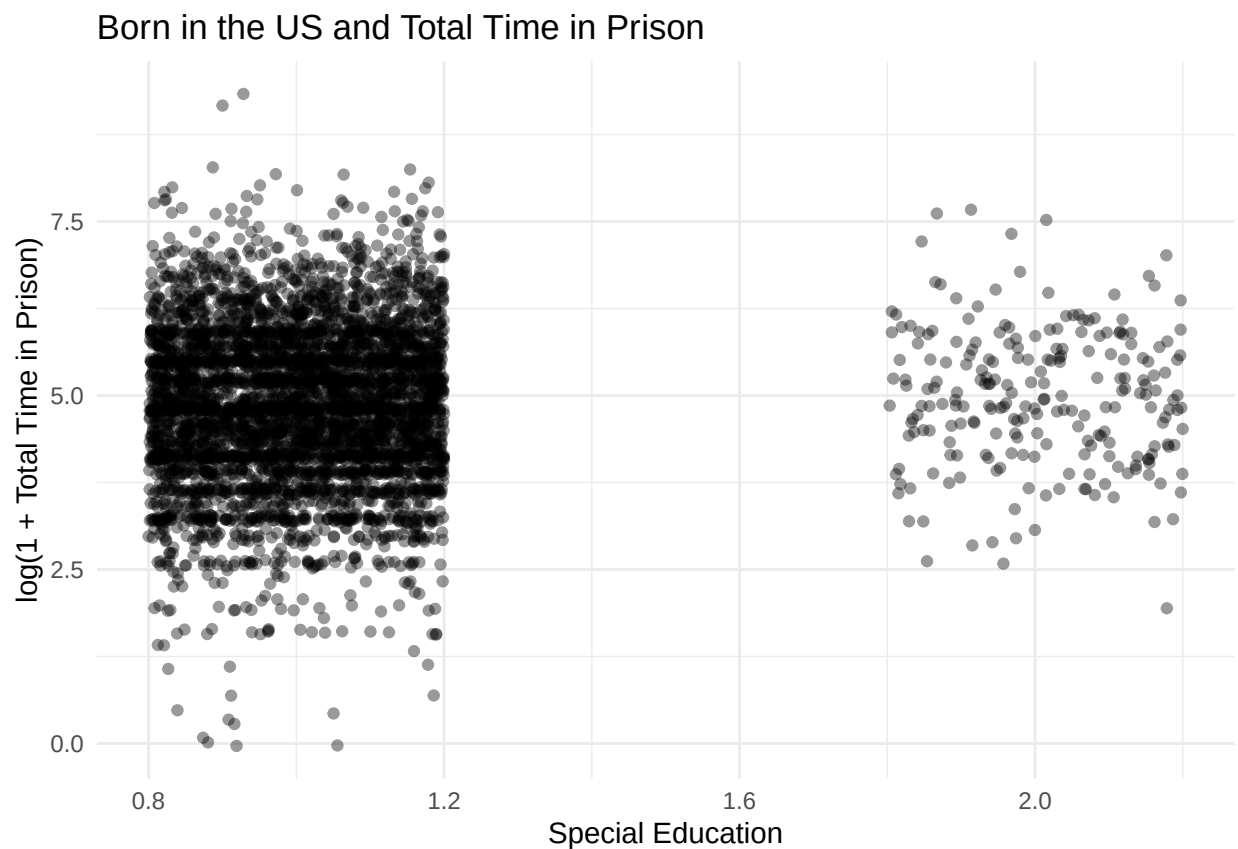
##	offense_num	Total	Max Sentence	Max Sentence	# of Years		
## NA	NA		NA		NA		
## NA.1	NA		NA		NA		
## NA.2	NA		NA		NA		
## NA.3	NA		NA		NA		
## NA.4	NA		NA		NA		
## NA.5	NA		NA		NA		
## NA.6	NA		NA		NA		
## NA.7	NA		NA		NA		
## NA.8	NA		NA		NA		
## NA.9	NA		NA		NA		
## NA.10	NA		NA		NA		
##	Max Sentence	# of Months	Max Sentence	# of Days	Years of Education	SpEd	
## NA		NA		NA	NA	NA	
## NA.1		NA		NA	NA	NA	
## NA.2		NA		NA	NA	NA	
## NA.3		NA		NA	NA	NA	
## NA.4		NA		NA	NA	NA	
## NA.5		NA		NA	NA	NA	
## NA.6		NA		NA	NA	NA	
## NA.7		NA		NA	NA	NA	
## NA.8		NA		NA	NA	NA	
## NA.9		NA		NA	NA	NA	
## NA.10		NA		NA	NA	NA	
##	Born in the US	Children	Foster Home	Homeless	Specify Home or Institution		
## NA	NA	NA	NA	NA	NA		
## NA.1	NA	NA	NA	NA	NA		
## NA.2	NA	NA	NA	NA	NA		
## NA.3	NA	NA	NA	NA	NA		
## NA.4	NA	NA	NA	NA	NA		
## NA.5	NA	NA	NA	NA	NA		
## NA.6	NA	NA	NA	NA	NA		
## NA.7	NA	NA	NA	NA	NA		
## NA.8	NA	NA	NA	NA	NA		
## NA.9	NA	NA	NA	NA	NA		
## NA.10	NA	NA	NA	NA	NA		
##	Lived with most time	V1264	V1252	Welfare	Public Housing	Father	Mother
## NA		NA	NA	NA	NA	NA	NA
## NA.1		NA	NA	NA	NA	NA	NA
## NA.2		NA	NA	NA	NA	NA	NA
## NA.3		NA	NA	NA	NA	NA	NA
## NA.4		NA	NA	NA	NA	NA	NA
## NA.5		NA	NA	NA	NA	NA	NA
## NA.6		NA	NA	NA	NA	NA	NA
## NA.7		NA	NA	NA	NA	NA	NA
## NA.8		NA	NA	NA	NA	NA	NA
## NA.9		NA	NA	NA	NA	NA	NA
## NA.10		NA	NA	NA	NA	NA	NA
##	Sibling	Tot_time	edu_group	offense_type			
## NA	NA	NA	<NA>	<NA>			
## NA.1	NA	NA	<NA>	<NA>			

```
## NA.2      NA      NA      <NA>      <NA>
## NA.3      NA      NA      <NA>      <NA>
## NA.4      NA      NA      <NA>      <NA>
## NA.5      NA      NA      <NA>      <NA>
## NA.6      NA      NA      <NA>      <NA>
## NA.7      NA      NA      <NA>      <NA>
## NA.8      NA      NA      <NA>      <NA>
## NA.9      NA      NA      <NA>      <NA>
## NA.10     NA      NA      <NA>      <NA>
```

```
ggplot(sblast2, aes(x = `Born in the US`, y = log1p(Tot_time))) +
  geom_jitter(width = 0.2, height = 0.05, alpha = 0.4) +
  labs(
    x = "Special Education",
    y = "log(1 + Total Time in Prison)",
    title = "Born in the US and Total Time in Prison"
  ) +
  theme_minimal()
```

```
## Warning in log1p(Tot_time): NaNs produced
```

```
## Warning: Removed 14 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
sblast2[sblast2$Children<0,] #19 observations missing for those who were born in the US or not
```

##	offense_num	Total	Max Sentence	Max Sentence	# of Years
## 303	1		1		60
## 1851	4		1		20
## 4900	1		1		14
## NA	NA		NA		NA
## 7583	1		1		6
## NA.1	NA		NA		NA
## NA.2	NA		NA		NA
## NA.3	NA		NA		NA
## 10204	1		1		14
## NA.4	NA		NA		NA
## NA.5	NA		NA		NA
## NA.6	NA		NA		NA
## NA.7	NA		NA		NA
## NA.8	NA		NA		NA
## NA.9	NA		NA		NA
## NA.10	NA		NA		NA
## 17288	1		1		70
## NA.11	NA		NA		NA
## NA.12	NA		NA		NA

##	Max Sentence	# of Months	Max Sentence	# of Days	Years of Education	SpEd
## 303		0		0	9	2
## 1851		0		0	9	2
## 4900		0		0	12	2
## NA		NA		NA	NA	NA
## 7583		0		0	8	1
## NA.1		NA		NA	NA	NA
## NA.2		NA		NA	NA	NA
## NA.3		NA		NA	NA	NA
## 10204		0		0	12	2
## NA.4		NA		NA	NA	NA
## NA.5		NA		NA	NA	NA
## NA.6		NA		NA	NA	NA
## NA.7		NA		NA	NA	NA
## NA.8		NA		NA	NA	NA
## NA.9		NA		NA	NA	NA
## NA.10		NA		NA	NA	NA
## 17288		0		0	12	1
## NA.11		NA		NA	NA	NA
## NA.12		NA		NA	NA	NA

##	Born in the US	Children	Foster Home	Homeless	Specify Home or Institution
## 303	1	-1	1	2	2
## 1851	1	-1	2	2	NA
## 4900	1	-2	2	2	NA
## NA	NA	NA	NA	NA	NA
## 7583	1	-1	1	1	2
## NA.1	NA	NA	NA	NA	NA
## NA.2	NA	NA	NA	NA	NA
## NA.3	NA	NA	NA	NA	NA
## 10204	1	-2	-2	-2	NA
## NA.4	NA	NA	NA	NA	NA

## NA.5	NA	NA	NA	NA	NA	NA
## NA.6	NA	NA	NA	NA	NA	NA
## NA.7	NA	NA	NA	NA	NA	NA
## NA.8	NA	NA	NA	NA	NA	NA
## NA.9	NA	NA	NA	NA	NA	NA
## NA.10	NA	NA	NA	NA	NA	NA
## 17288	1	-1	2	2	NA	NA
## NA.11	NA	NA	NA	NA	NA	NA
## NA.12	NA	NA	NA	NA	NA	NA
##	Lived with most time	V1264	V1252	Welfare	Public Housing	Father Mother
## 303	12	17	12	1	2	-1 1
## 1851	2	-1	18	2	2	2 2
## 4900	2	NA	NA	2	2	2 2
## NA	NA	NA	NA	NA	NA	NA NA
## 7583	9	16	-2	1	1	1 -1
## NA.1	NA	NA	NA	NA	NA	NA NA
## NA.2	NA	NA	NA	NA	NA	NA NA
## NA.3	NA	NA	NA	NA	NA	NA NA
## 10204	-2	NA	-2	-2	-2	-2 -2
## NA.4	NA	NA	NA	NA	NA	NA NA
## NA.5	NA	NA	NA	NA	NA	NA NA
## NA.6	NA	NA	NA	NA	NA	NA NA
## NA.7	NA	NA	NA	NA	NA	NA NA
## NA.8	NA	NA	NA	NA	NA	NA NA
## NA.9	NA	NA	NA	NA	NA	NA NA
## NA.10	NA	NA	NA	NA	NA	NA NA
## 17288	1	12	NA	1	2	2 2
## NA.11	NA	NA	NA	NA	NA	NA NA
## NA.12	NA	NA	NA	NA	NA	NA NA
##	Sibling	Tot_time	edu_group	offense_type		
## 303	1	730.0000	< HS	Violent		
## 1851	1	243.3333	< HS	Public Order		
## 4900	2	170.3333	HS/GED	Violent		
## NA	NA	NA	<NA>	<NA>		
## 7583	2	73.0000	< HS	Violent		
## NA.1	NA	NA	<NA>	<NA>		
## NA.2	NA	NA	<NA>	<NA>		
## NA.3	NA	NA	<NA>	<NA>		
## 10204	-2	170.3333	HS/GED	Violent		
## NA.4	NA	NA	<NA>	<NA>		
## NA.5	NA	NA	<NA>	<NA>		
## NA.6	NA	NA	<NA>	<NA>		
## NA.7	NA	NA	<NA>	<NA>		
## NA.8	NA	NA	<NA>	<NA>		
## NA.9	NA	NA	<NA>	<NA>		
## NA.10	NA	NA	<NA>	<NA>		
## 17288	2	851.6667	HS/GED	Violent		
## NA.11	NA	NA	<NA>	<NA>		
## NA.12	NA	NA	<NA>	<NA>		

```
ggplot(sblast2, aes(x = Children, y = log1p(Tot_time))) +
  geom_jitter(width = 0.2, height = 0.05, alpha = 0.4) +
  geom_smooth(method = "loess", se = TRUE) +
  labs(
```

```

    x = "Special Education",
    y = "log(1 + Total Time in Prison)",
    title = "Born in the US and Total Time in Prison"
) +
theme_minimal()

## Warning in log1p(Tot_time): NaNs produced
## Warning in log1p(Tot_time): NaNs produced

## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 16 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : pseudoinverse used at 2.02

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : neighborhood radius 1.02

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : reciprocal condition number 2.5923e-29

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = parametric,
## : There are other near singularities as well. 1

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : pseudoinverse used at
## 2.02

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : neighborhood radius
## 1.02

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : reciprocal condition
## number 2.5923e-29

## Warning in predLoess(object$y, object$x, newx = if (is.null(newdata)) object$x
## else if (is.data.frame(newdata))
## as.matrix(model.frame(delete.response(terms(object))), : There are other near
## singularities as well. 1

## Warning: Removed 16 rows containing missing values or values outside the scale range
## ('geom_point()').

```

Born in the US and Total Time in Prison

